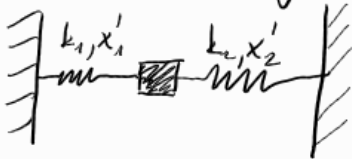


- makroskopske opazljivke $p, V, T, C_V, \beta, \chi_T, \dots$

- termodinamični potencial U, S, H, F, G
 $\nearrow$ notranja energija $\nearrow$ prosta energija $\nearrow$ prosta entalpija

TD ravnovesje

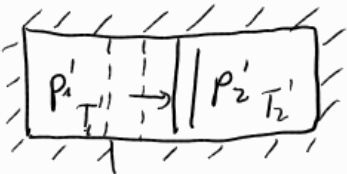
- mehanično ravnovesje



$$|F_1| = |F_2|$$

$\rightarrow$ da sistem pride v ravnovesje, morajo vzeti izmenjati delo

$$|k_1(x_1 - x_1')| = |k_2(x_2 - x_1')|$$

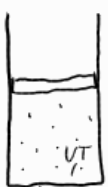


but $P_1 = P_2 \rightarrow$ izmenjava dela

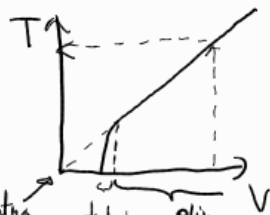
$T_1 = T_2 \rightarrow$ izmenjava toplote

Temperatura

- operativna definicija: plinski termometer



$p = \text{konst.}$



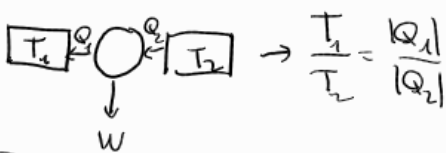
absolutna ničla 0 K
 trdnina, plin
 kapljavina

Guy-Lussac:

$$\frac{V}{T} = \frac{V'}{T'} \quad p = \text{konst.}$$

- trojna točka vode $\rightarrow 273,16 \text{ K}$

- prek Carnotovega topl. stroja:



$$T_2 > T_1$$

TD spremenljivke

$$T; p, V \rightarrow dW = F ds = -p \cdot S ds = -p dV$$

$$T; U, e \rightarrow dW = U de$$

$$T; \vec{A}, \vec{p}_m = \mu_0 V \vec{M} \rightarrow dW = \mu_0 V H d\vec{M}$$

$$\rightarrow dW = \gamma dx$$

posplošena
sila;
intenzivna
spremenljivka

(če sta sist.
v rav. $\rightarrow \gamma_1 = \gamma_2$)

posplošeni
odziv;
ekstenzivna
spremenljivka

($X = X_1 + X_2$) $\rightarrow$ npr. celoten $V = V_1 + V_2$

Idealni plin:

Boyleov zakon: $pV = p'V' \quad T = \text{konst.}$

Gay-Lussacov zakon: $V/T = V'/T' \quad p = \text{konst.}$

- enačba stanja idealnega plina

$$\frac{pV}{T} = \frac{mR}{M} = \text{konst.} \quad \text{INTEGRALSKA OBLIKA}$$

$$\frac{dV}{V} = \dots dT + \dots dp \quad \text{DIFERENCIALNA OBLIKA}$$

$$V = V(p, T)$$

$$\frac{dV}{V} = \underbrace{\frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p}_{\beta} dT - \underbrace{\left[-\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T \right]}_{\chi_T} dp$$

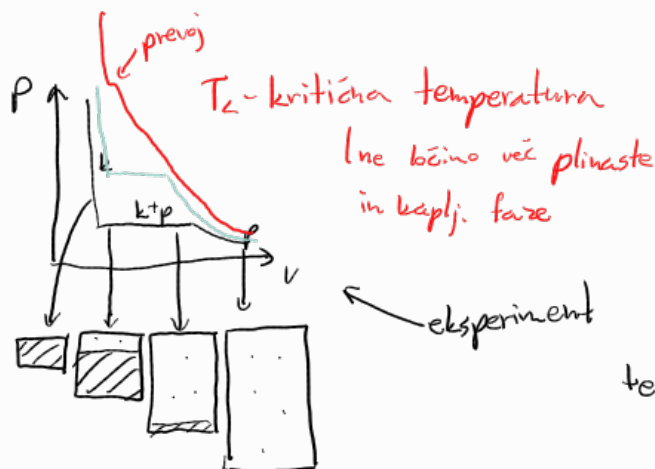
$\rightarrow$ za idealni plin: $\beta = \frac{1}{T}$

$$\chi_T = \frac{1}{p}$$



Realni plin

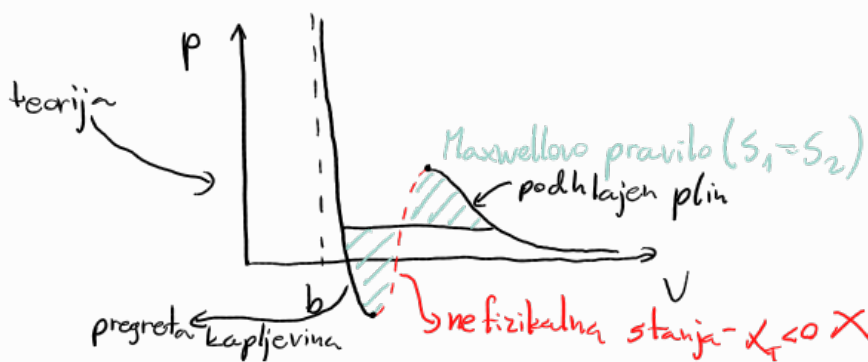
- fazni prehod kapljevine-plin



Van der Waalsova enačba stanja

$$\left(p + \frac{a}{V_m^2}\right)(V_m - b) = RT \quad V_m = \frac{MV}{m}$$

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$



Zgledi: enačbe stanj

1) paramagnet

- $M = \chi H$
↑ magnetna susceptibilnost

$$C = \chi T$$

$$\boxed{M = \frac{C}{T} H}$$

Curiejev
zakon
(za velike T)

diamagnet $\chi < 0$
 $|\chi| \ll 1$

{ paramagnet $\chi > 0$
 $|\chi| \ll 1$

feromagnet $\chi \gg 1$

2) Superprevodnik

- Meissnerjev pojav: $B = 0 = \mu_0 (\underbrace{\vec{M} + \vec{H}}_{=0})$

$$\vec{M} = -\vec{H}$$

za nizke T

3) Sevanje črnega telesa

$$p = \frac{4}{3} \frac{\sigma}{c} T^4$$